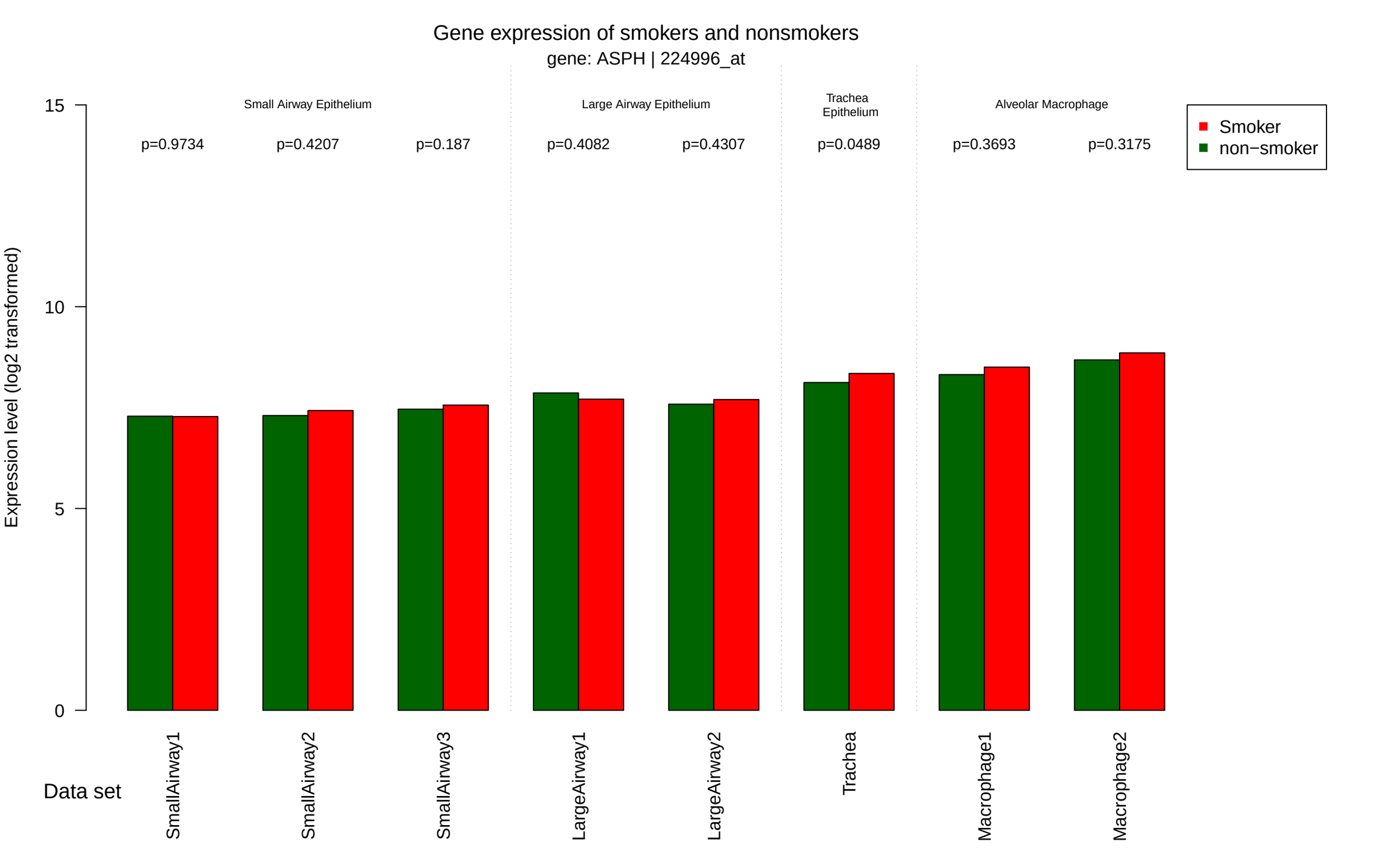
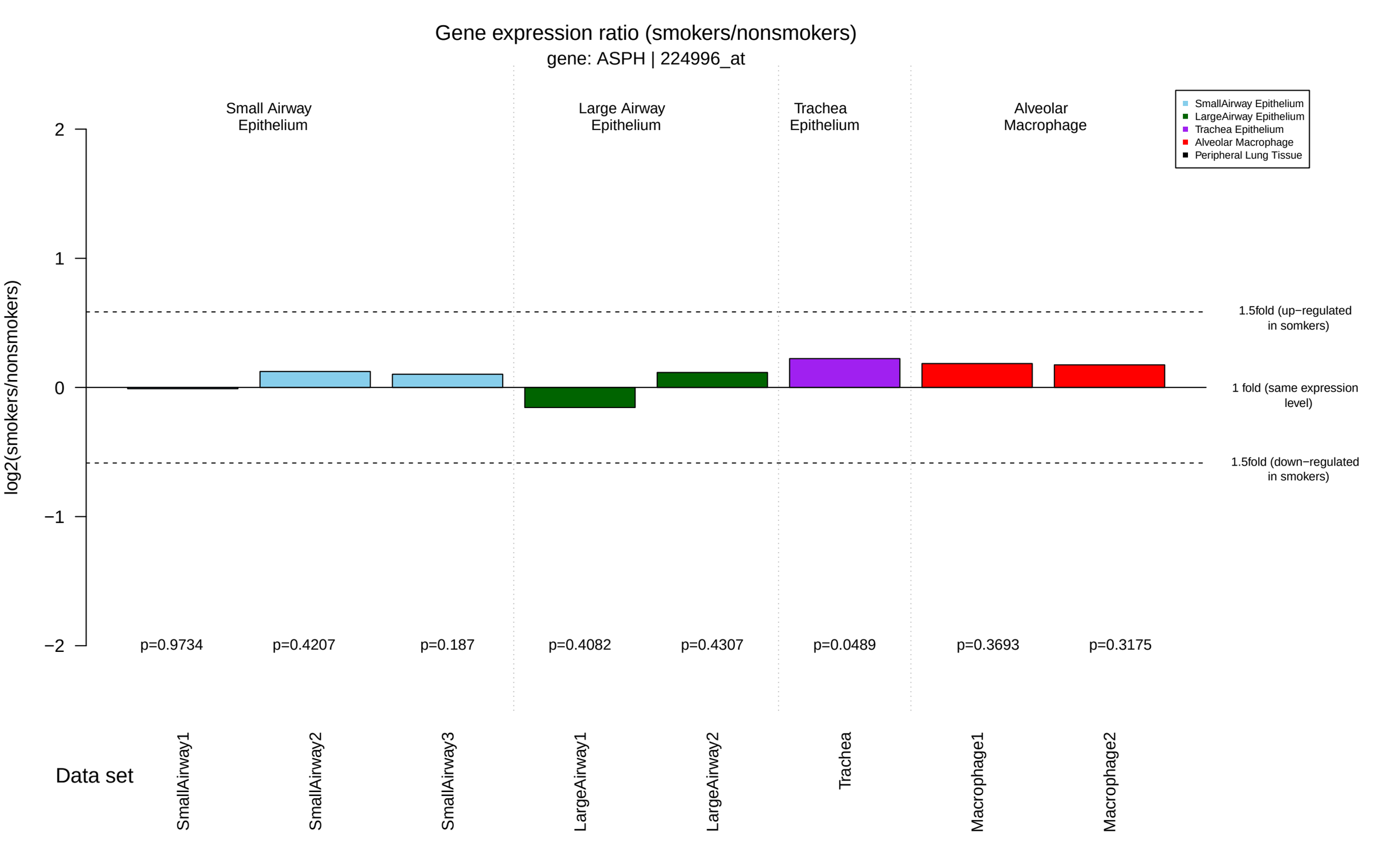
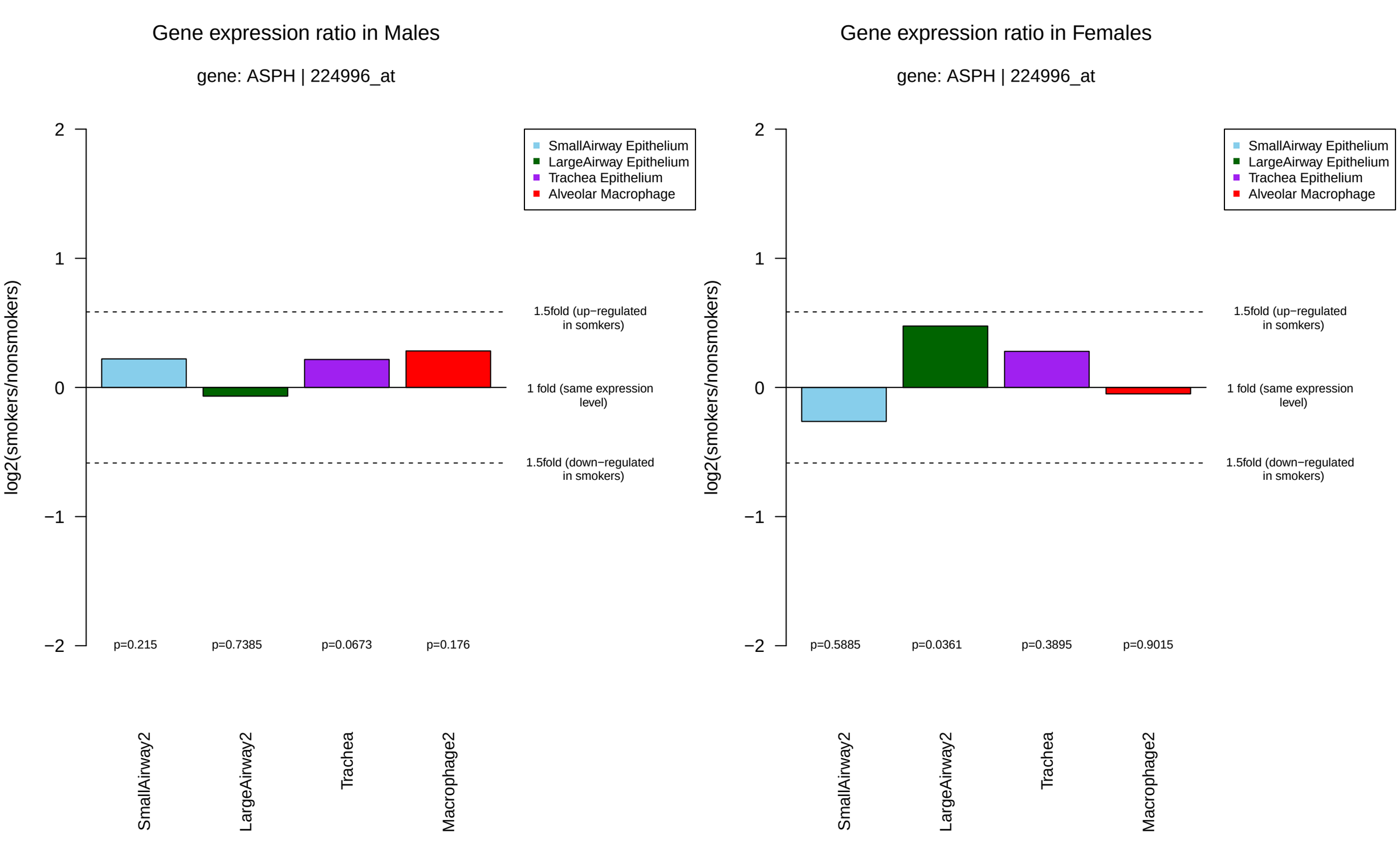
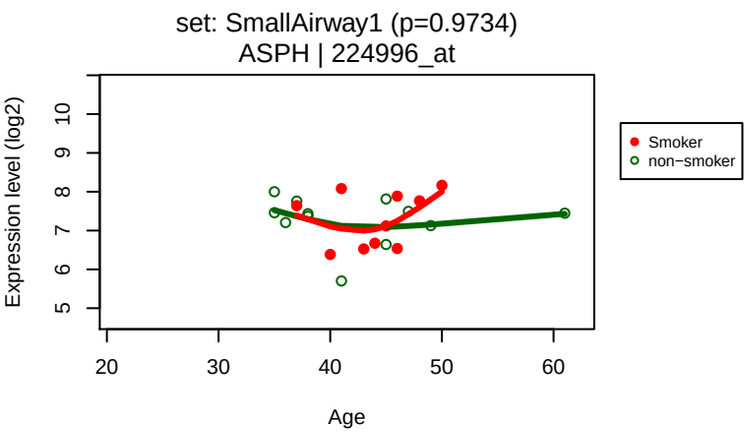
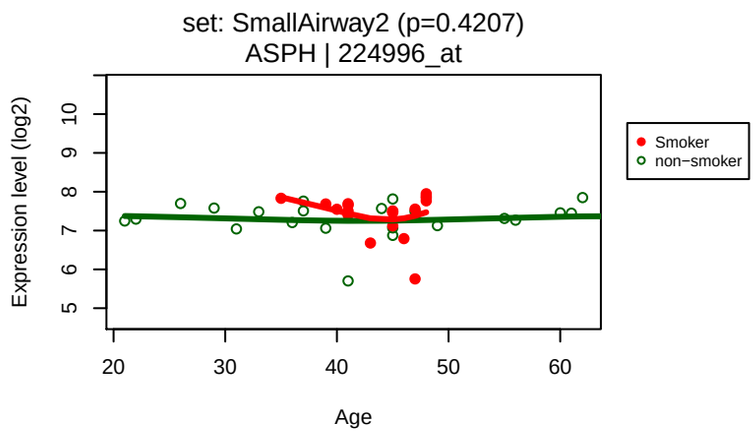
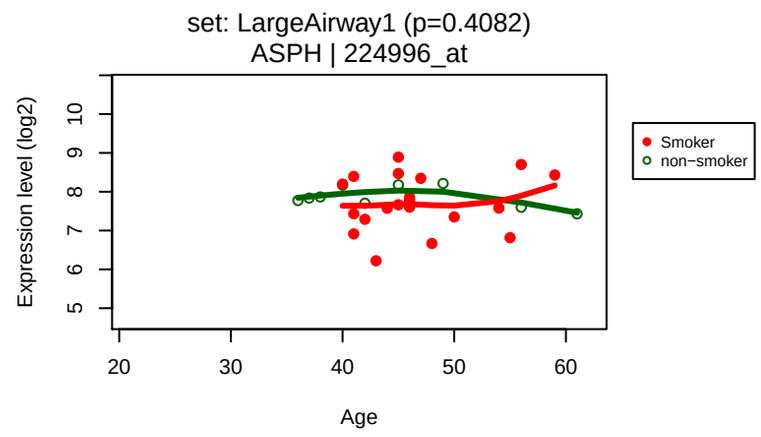

Gene expression of smokers and nonsmokers
gene: ASPH | 224996_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9734
p=0.4207
p=0.187
p=0.4082
p=0.4307
p=0.0489
p=0.3693
p=0.3175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

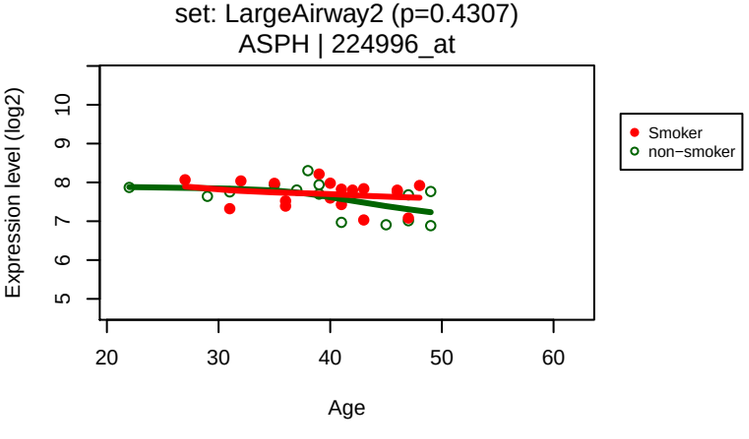

Gene expression ratio (smokers/nonsmokers)
gene: ASPH | 224996_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9734
p=0.4207
p=0.187
p=0.4082
p=0.4307
p=0.0489
p=0.3693
p=0.3175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

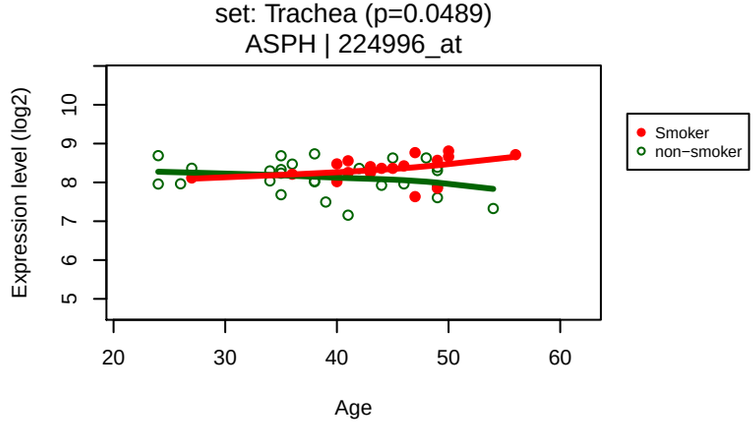

Gene expression ratio in Males
gene: ASPH | 224996_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.215
p=0.7385
p=0.0673
p=0.176
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPH | 224996_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5885
p=0.0361
p=0.3895
p=0.9015
SmallAirway2
LargeAirway2
Trachea
Macrophage2

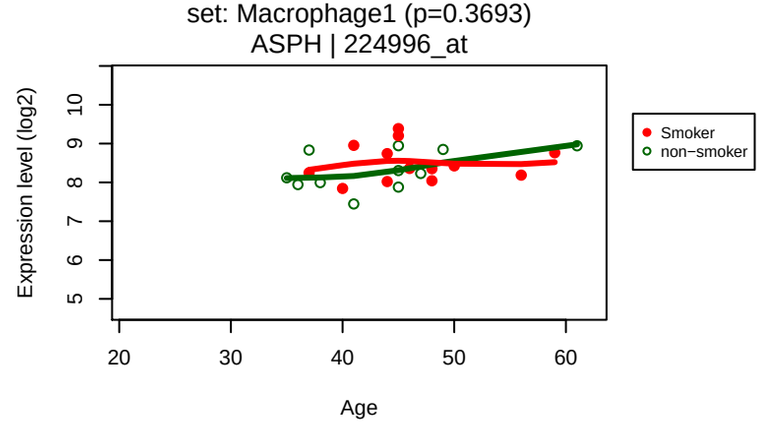

set: SmallAirway1 (p=0.9734)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker

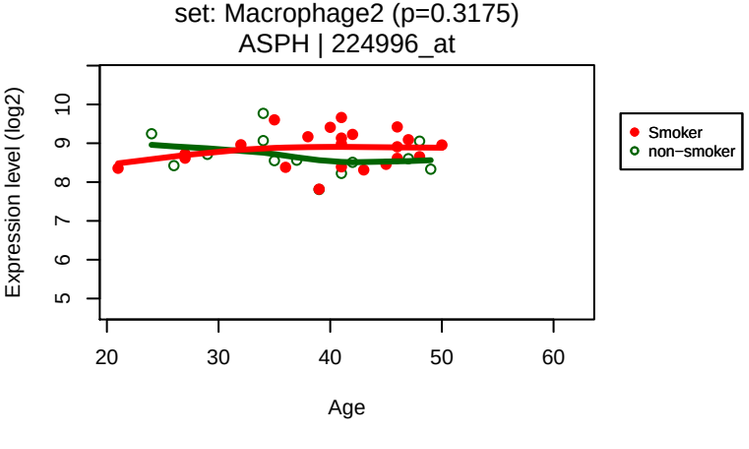

set: SmallAirway2 (p=0.4207)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker

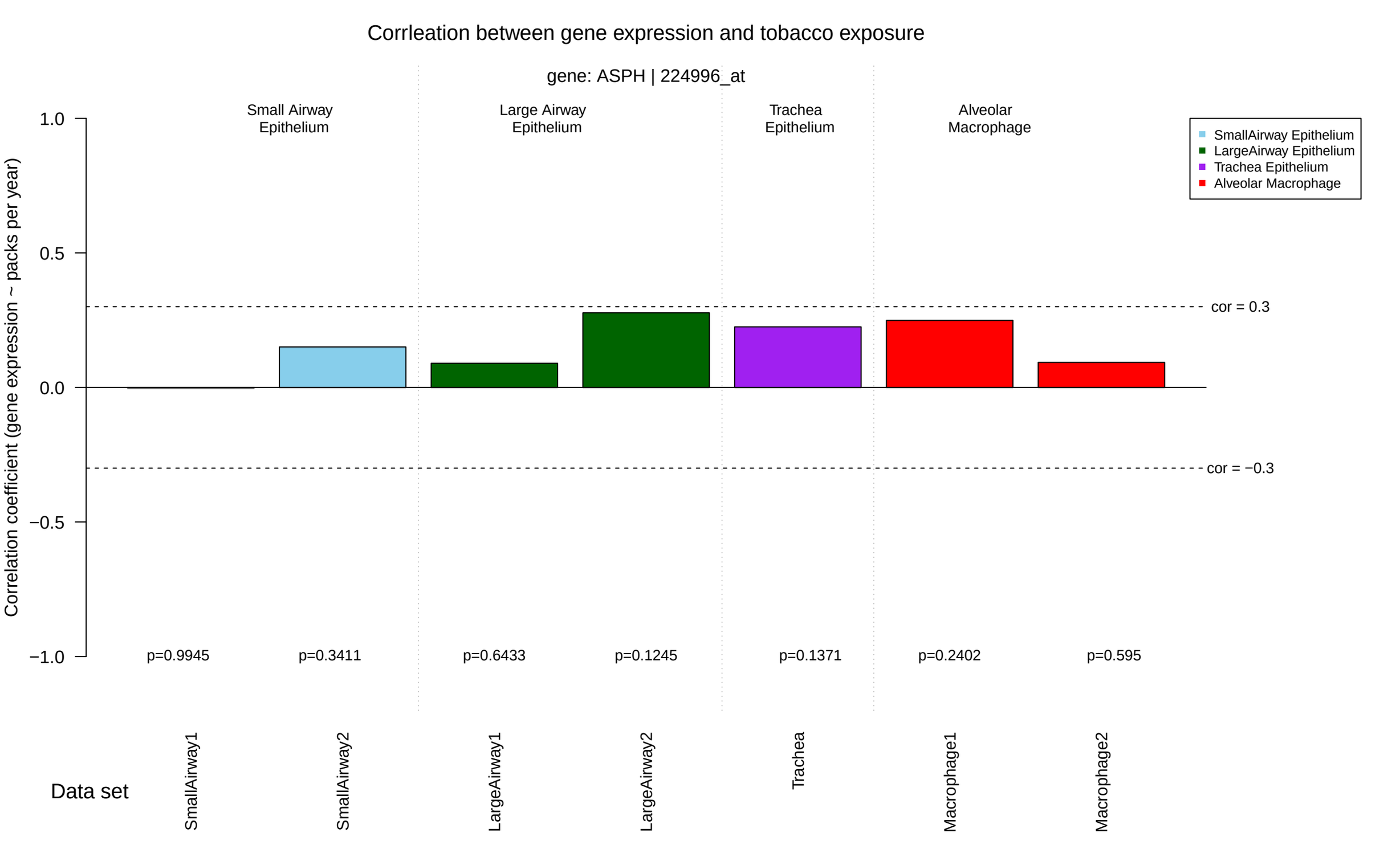

set: LargeAirway1 (p=0.4082)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4307)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0489)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3693)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3175)
ASPH | 224996_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPH | 224996_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9945
p=0.3411
p=0.6433
p=0.1245
p=0.1371
p=0.2402
p=0.595
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2